

Answer Key — Biology Grade 10

Sub-Strand 1.2: Chemicals of Life • Lesson 1: Predict: What Is Food Really Made Of?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	C	A	A	D	A	A

USE AFTER: PREDICT PHASE — Notice & Wonder entry after silent observation of Photo A and Photo B

Q1. Two children live in the same village near Kisumu and eat about the same amount of food each day. Child A has a swollen belly, thin limbs and reddish hair. Child B looks healthy and energetic. Which of these is something you NOTICE (an observation) rather than a guess?

- A. Child B has more money than Child A's family.
- B. Child A was born with a disease.
- C. Child A is sick because something is missing from the food.
- D. Child A has a swollen belly and thin limbs.**

Correct answer: D. An observation is something you can see directly, such as a swollen belly and thin limbs. The other statements are guesses about causes that the photographs do not show.

USE AFTER: PREDICT PHASE — Teacher reveals the food details (ugali vs githeri) and students write 'I think the difference in the children's bodies is caused by ___ because ___'

Q2. Child A eats ugali (maize starch) as the main meal every day. Child B eats githeri (maize and beans) regularly. Which statement matches the information given about the two children?

- A. Both children eat only ugali every day.
- B. The children live in different parts of Kenya.
- C. Both children eat about the same amount of food and live in the same village near Kisumu.**
- D. Child B eats much more food each day than Child A.

Correct answer: C. The caption states that both children eat approximately the same amount of food and live in the same village. What differs is the kind of food, not the quantity or the place.

USE AFTER: OBSERVE PHASE — Groups complete the two-column table 'What the evidence shows us' vs 'What we still need to find out' after reading the Phenomenon Card

Q3. A group is filling in a two-column table about the two Kisumu children: 'What the evidence shows us' and 'What we still need to find out'. Which entry belongs in the SECOND column?

- A. Which specific chemicals are different between ugali and githeri.**
- B. Child B eats githeri and looks healthy and energetic.
- C. Both children live in the same village near Kisumu.
- D. Child A eats ugali every day and has a swollen belly.

Correct answer: A. The first three entries are facts already given on the Phenomenon Card. The chemical difference between the foods has not been investigated yet, so it belongs under what we still need to find out.

USE AFTER: OBSERVE PHASE — Debrief of the Phenomenon Card: 'What does this evidence show us, and what does it NOT tell us yet?'

Q4. Right now the class has only seen photographs and read a short card about the two children. Which is the best response to the question 'Why do the two children have such different bodies?'

- A. We do not know yet. We need to find out what the two foods are made of before we can explain it.**
- B. The child with the swollen belly is just tired from playing too much.
- C. Ugali is bad food and githeri is good food, so nothing more needs to be studied.
- D. Child A is not eating enough food, and that is the only reason.

Correct answer: A. At this stage the class has evidence of a puzzle but has not investigated the foods. Saying honestly that it is not yet explained keeps the question open. The other answers are unsupported or contradict the given information that the food quantity is similar.

USE AFTER: EXPLAIN PHASE — Chemical Families Card Sort: the 6 'Chemical Family' cards

Q5. In the Chemical Families Card Sort, groups match foods such as ugali, githeri, sukuma wiki, nyama choma, fish from Lake Victoria and mandazi to the chemical family each is richest in. Which of these lists names the six chemical families used in the card sort?

- A. Carbohydrates, proteins, lipids, vitamins, mineral salts, oxygen
- B. Starch, ugali, beans, oil, salt, water
- C. Carbohydrates, proteins, lipids, vitamins, fibre, air
- D. Carbohydrates, proteins, lipids, vitamins, mineral salts, water**

Correct answer: D. The six chemicals of life named in this lesson are carbohydrates, proteins, lipids (fats and oils), vitamins, mineral salts and water. The other lists include foods, air, oxygen or fibre, which are not among the six families.

USE AFTER: EXPLAIN PHASE — Whole-class debrief after the gallery walk and revision of the card sort

Q6. After the card sort and gallery walk, some groups disagree about which chemical family a food is richest in, and nobody corrects them. How will the class find out which matches are correct?

- A. By doing chemical tests on foods in the next lessons.**
- B. By choosing the match that sounds most scientific.
- C. By asking the group with the neatest handwriting.
- D. By voting for the match that most groups chose.

Correct answer: A. The lesson says no tests are done today and that later lessons will use chemical tests to check the matches. Voting or sounding scientific does not provide evidence.

USE AFTER: MODEL BUILDING PHASE — Students draw their first explanatory model and complete the 'Model Limitations Statement'

Q7. You have drawn your first model of what happens inside the body of each child, and written 'My model cannot yet explain ____ because ____'. Which statement about this first model is correct?

- A. It is your best current explanation and will be revised as you learn more.**
- B. It should be erased if it turns out to be wrong.
- C. It must use all the correct chemical names before it is finished.
- D. It is your final answer for the whole unit.

Correct answer: A. A first model is a best current explanation that can use everyday words. Students keep it without erasing and revise it lesson by lesson, so being incomplete is expected.